Supplementary Materials for Controlling the Perceived Competitive Balance in Sports Leagues Through Scheduling

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Abstract

This paper investigates how scheduling influences the perceived competitive balance in sports leagues. By analyzing 1,236 seasons across 172 professional leagues in basketball, handball, soccer, and volleyball, we observe that a tournament's behaviour eventually diverges from what would be expected from perfectly balanced tournaments. We propose novel heuristic algorithms that leverage team rankings to control this divergence. Our findings demonstrate that these heuristics can significantly extend the period of perceived competitive balance in a large majority of the tournaments analyzed, potentially enhancing audience engagement and complementing existing strategies for maintaining interest throughout the season.

Keywords: Analytics, Metaheuristics, OR in sports, Scheduling

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Appendix A. Other Point System

Since the *perceived competitive balance* depends on the tournament standings, the point system directly affects its temporal progression. For most competitions, 3 points for a win, 1 for a draw, and 0 for a loss is the standard choice. However, for volleyball, another point system, based on the number of sets won, is commonly used. In this system, a win by 3 sets to 0, or 3 sets to 1 still awards 3 points to the winner and 0 to the loser, whereas a win by 3 sets to 2 would only result in 2 points to the winner, and 1 to the loser, since it is natural to assume that in such a close match both teams are closely matched.

Figure A.1 shows the comparison between the two systems. The main difference can be attributed to the (2,1)-system attempting to maintain similarly skilled teams closer together by design. Consequently, only giving a 1-point difference (2 winner, 1 loser) in the standings, rather than a 3-point one, prevents a close win from disproportionally affecting the point distribution. However, the effect on τ is minimal, with the change in the point system mainly affecting teams in the middle of the standings, since the best (worst) team would likely still win (lose) most matches by 3 sets to 0, or 3 sets to 1. This bias is not present when estimating the quantile envelope in our simulations, so best and worst teams do not distinguish themselves from the rest as fast as seen in reality. As a result, *perceived competitive balance* for the (2,1)-system tends to be slightly smaller than what is observed for the (3,0)-system, as seen in the boxplots.

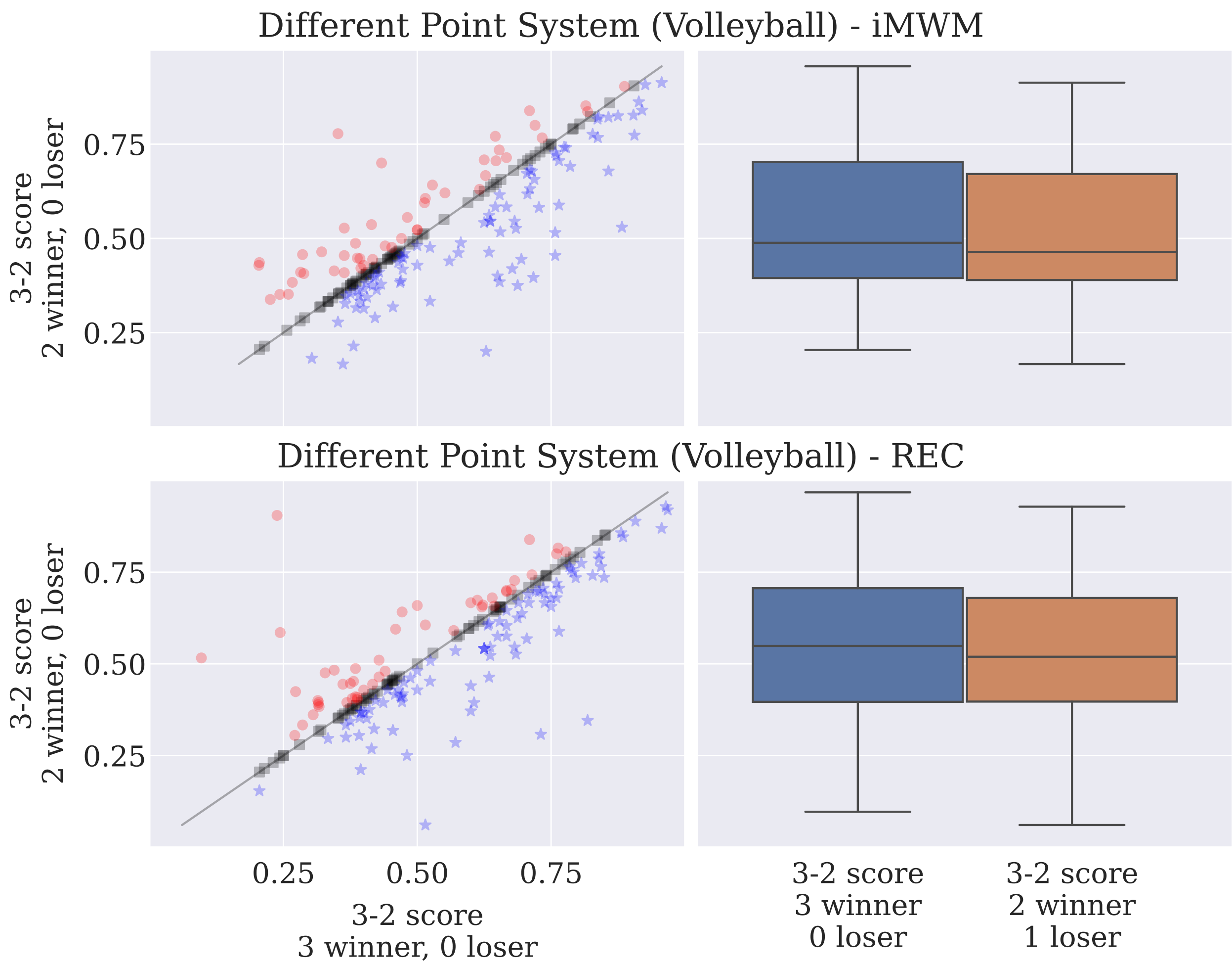

Figure A.1: **Another point system.** Comparison of *oracle* **PCB** between different point systems for our heuristics. In the first point system, a win always grants 3 points for the winner and 0 for the loser, whereas in the second one, a 3 sets to 2 win grants 2 points for the winner and 1 for the loser.

Appendix B. Other Metrics

Competitive balance in sports leagues assesses the uniformity in the skill distribution among teams. It is a critical concept because balanced tournaments, as per the *uncertainty of outcome* hypothesis (Rottenberg, 1956), are more unpredictable, which generates more interest from fans and increases match viewership (Neale, 1964; Jennett, 1984; Fort and Quirk, 1995; Dobson and Goddard, 2001; Forrest and Simmons, 2002; Borland, 2003; Szymanski, 2003). One caveat is that the teams' actual skill levels are typically unknown, so competitive balance is instead inferred from observed tournament data and measured within a particular season or across multiple seasons of a given league (Szymanski, 2003; Buzzacchi et al., 2010; Gerrard and Kringstad, 2022).

Within a season, the most common approach is to use *dispersion* and *inequality* metrics on the distribution of points and victories in the standings (Quirk and Fort, 1997; Zimbalist, 2002; Humphreys, 2002; Owen, 2014; Michie and Oughton, 2004; Doria and Nalebuff, 2021; Manasis et al., 2022). The standard deviation (Scully, 1989; Quirk and Fort, 1997) and the Gini coefficient (Quirk and Fort, 1997; Schmidt and Berri, 2001), when applied to the percentage of matches won by each team, are examples of simple and widely used metrics for measuring competitive balance within a season. The lower the difference in points between the best and worst teams, the more balanced the tournament is. For competitive balance across seasons, rank correlation and tail-outcome persistence measures are used to characterize how league outcomes in one season are replicated in the next (Daly and Moore, 1981; Maxcy, 2002; Gerrard and Kringstad, 2023). For instance, more balanced leagues tend to have different champions and title contenders across their seasons.

Although the standings' variance is one available option, other metrics have been more favored in recent times. As a result, using the variance during the formulation (Section 3) might seem unorthodox. In this section, we show that choosing other metrics of competitive balance would not significantly change our results.

The first two metrics we analyze in this section are related to the Herfindahl–Hirschman Index (HHI), which, in sports leagues, compares the fraction of points each team has compared to the total. Consider tournaments composed of N teams and T rounds. Following our previous notation, $X_{i,t}$ is the points accrued by team i up to round t. Similarly, X_t and $\overline{X_t}$ are, respectively, the total and the average number of points up to round t. Then, mathematically, the HHI is given by:

$$\mathrm{HHI} = \sum_{i=1}^{N} \left(\frac{X_{i,T}}{X_T} \right)^2. \tag{B.1}$$

However, in this definition, factors such as the number of teams in the tournament influences its value. To mitigate this influence, the metric we consider is common normalization by the expected bounds seen in tournaments leagues:

$$\mathrm{nHHI} = \frac{\mathrm{HHI} - \frac{1}{n}}{\mathrm{UB_{HHI}} - \frac{1}{\mathrm{n}}}, \tag{B.2}$$

where $\frac{1}{n}$ is the HHI for a perfect balanced tournament, and UB is the upper bound for sports leagues, estimated from the most imbalanced tournament (Owen et al., 2007; Fort and Quirk, 1997; Utt and Fort, 2002). It can be constructed following a simple rule: the team A that plays the most matches wins all of them, the team B that plays the second-largest number of matches wins all except against the team A, and so forth. In double round-robin tournaments, where all teams play the same amount, the UB can be analytically calculated (Owen et al., 2007), whereas, in less structured scenarios, it is easier to build the most imbalanced tournament and estimate the UB directly from it.

The second metric, is the Herfindahl Index of Competitive Balance (HICB). It also controls for the number of teams in its estimation, approximating the HHI to a variance measure (Doria and Nalebuff, 2021).

$$\mathrm{HICB} = \frac{\mathrm{HHI}}{\frac{1}{n}}. \tag{B.3}$$

The last metric we will consider in this analysis is the Gini index. In the context of sports leagues, the Gini coefficient measures how far the point distribution deviates from the equality distribution. Its usual definition is geometrically tied to the Lorentz curve, but it is algebraically given by

$$\mathrm{Gini} = \frac{1}{2\overline{X_T} n^2} \sum_{i=1}^{N} \sum_{j=1}^{N} |s_i - s_j|. \tag{B.4}$$

However, such expression assumes an upper bound that cannot be reached in sports tournaments; thus, similar to the nHHI, we normalize the Gini coefficient by the upper bound obtained from the

Table B.1: Correlation between the oracle perceived competitive balance τ-`maximizer` using variance and what would be obtained if we had used other popular competitive balance metrics.

	nHHI		HICB		nGini	
	iMWM	**REC**	**iMWM**	**REC**	**iMWM**	**REC**
Basketball	0.994	0.904	0.994	0.904	0.967	0.887
Handball	0.944	0.924	0.944	0.924	0.935	0.905
Soccer	0.908	0.871	0.908	0.871	0.879	0.850
Volleyball	0.928	0.874	0.928	0.875	0.908	0.842

most imbalanced tournament (Utt and Fort, 2002):

$$\text{nGini} = \frac{\text{Gini}}{\text{UB}_{\text{Gini}}}. \tag{B.5}$$

In Table B.1 we present the correlation between the *perceived competitive balance* calculated using *variance* and the same coefficient calculated using the competitive imbalance metrics described above. We only show our results for the oracle approach, since it is almost for all other kinds of **PCB**. The correlation in the table is almost perfect, strongly suggesting that the results would be equivalent had we used other competitive balance metrics instead of the variance. Choosing the variance for the main text was mainly a matter of its simplicity.

Appendix C. Break Minimization

One the most fundamental constraints in tournament scheduling is break-minimization, that is, minimizing the number of times a team play at home (or away) in two consecutive rounds. It guarantees that all teams will consistently be able to play as the home team, dispersing the effects of home-court advantage throughout the tournament and allowing local fans to support their team.

In our **REC** methodology, the rotations in our *Matching step* (Algorithm 3) already favor a break-minization procedure. As long as we alternate where teams play, rather than force the best team to play at home, each *Matching Step* would automatically minimize the number of breaks. Consequently, the would only occur between the end of one *Matching Step* and the start of another. Unfortunately, given the greed nature of our **iMWM**, adding break-minimization during the algorithm is not as simple. Therefore, for both approaches, we apply the same procedure: minimizing breaks *given* the optimal schedule.

Consider a single round-robin tournament $\boldsymbol{\mathcal{S}}_{t,i,j}$ with N teams and T rounds. Let $h_{t,i}$ be a variable indicating if a home break for team i happened between rounds $t-1$ and t; and let $a_{t,i}$ be the equivalent for away breaks. In this scenario, minimizing the number of breaks is represented by the following integer-programming condition:

$$\min \sum_{t=2}^{T} \sum_{i=1}^{N} \Big(h_{t,i} + a_{t,i} \Big) \tag{C.1}$$

If $\boldsymbol{\mathcal{S}}^*$ is the optimal schedule generated by **REC** or **iMWM**, then its breaks can be minimized by imposing the following constraints:

$$\boldsymbol{\mathcal{S}}_{t,i,j} + \boldsymbol{\mathcal{S}}_{t,j,i} = \boldsymbol{\mathcal{S}}^*_{t,i,j} + \boldsymbol{\mathcal{S}}^*_{t,j,i} \qquad \forall\ (i,j) \in \{1,\ \cdots,\ N\}^2 \quad t \in \{1,\ \cdots,\ T\} \tag{C.2}$$

$$\sum_{j=1}^{N} \Big(\boldsymbol{\mathcal{S}}_{t-1,i,j} + \boldsymbol{\mathcal{S}}_{t,i,j} \Big) \leq 1 + h_{t,i} \qquad \forall\ i \in \{1,\ \cdots,\ N\} \quad t \in \{2,\ \cdots,\ T\} \tag{C.3}$$

$$\sum_{j=1}^{N} \Big(\boldsymbol{\mathcal{S}}_{t-1,j,i} + \boldsymbol{\mathcal{S}}_{t,j,i} \Big) \leq 1 + a_{t,i} \qquad \forall\ i \in \{1,\ \cdots,\ N\} \quad t \in \{2,\ \cdots,\ T\} \tag{C.4}$$

Equation C.2 guarantees that whatever round our heuristics placed the (i, j) match in $\boldsymbol{\mathcal{S}}^*$ will be kept in the break-minimized schedule. Since we are dealing with a *single* round-robin schedule, only one of $\boldsymbol{\mathcal{S}}^*_{t,i,j}$ and $\boldsymbol{\mathcal{S}}^*_{t,j,i}$ can be equal to one. Thus, the right-hand side will be equal to one if the match (i, j) happened in round t, and zero otherwise. Equation C.3 and Equation C.4 leverage the indicator variables $h_{t,i}$ and $a_{t,i}$ to allow for a break to happen in round t if necessary. For instance, when a home-break happens in round t, the left-hand sum must be equal to two, forcing $h_{i,j}$ to be 1. As a result, this break is accounted for in our optimization condition.

It is important to note, however, that only the first constraint enforces our heuristic's schedule to be used. The other constraints and the minimization condition are well-known integer-programming artifacts to minimize the number of breaks (Briskorn, 2008). Finding schedules using these artifacts by themselves can be a computationally expensive procedure; however, imposing our schedule as a initial condition meaningfully reduces the search space, allowing an optimal solution to be found even if N is large.

Appendix D. Results Segregated by Sport

Results separated by sport. The label indicates the type of coefficient:

Letter: O for Oracle, and Y for yesterday model

Subscript: Worst team as home ( ), and break-minimizer ($BreakMin$)

Superscript: Maximizer (Max), and Half-maximizer ($MaxMin$)

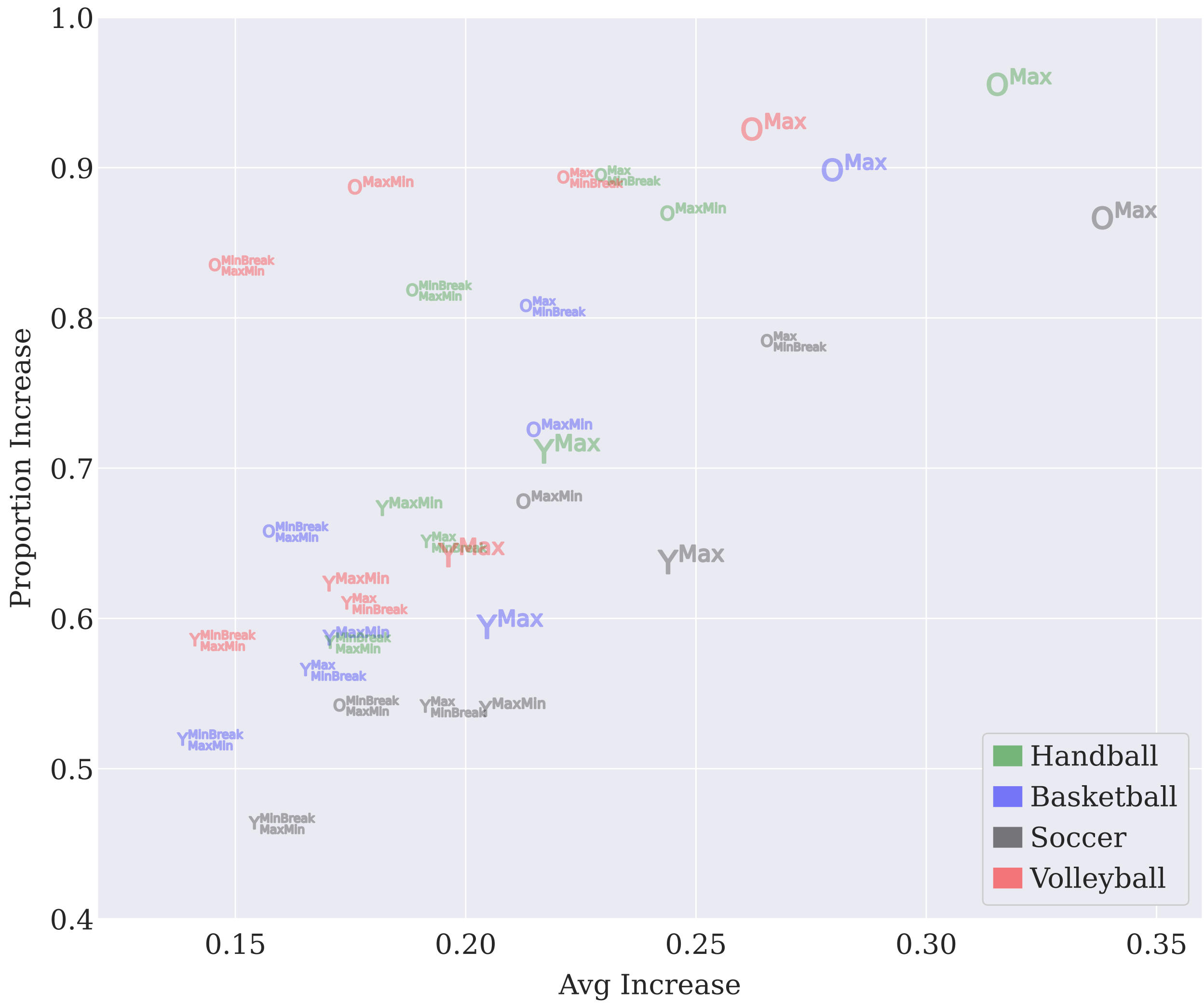

Figure D.2: Results for **iMWM**.

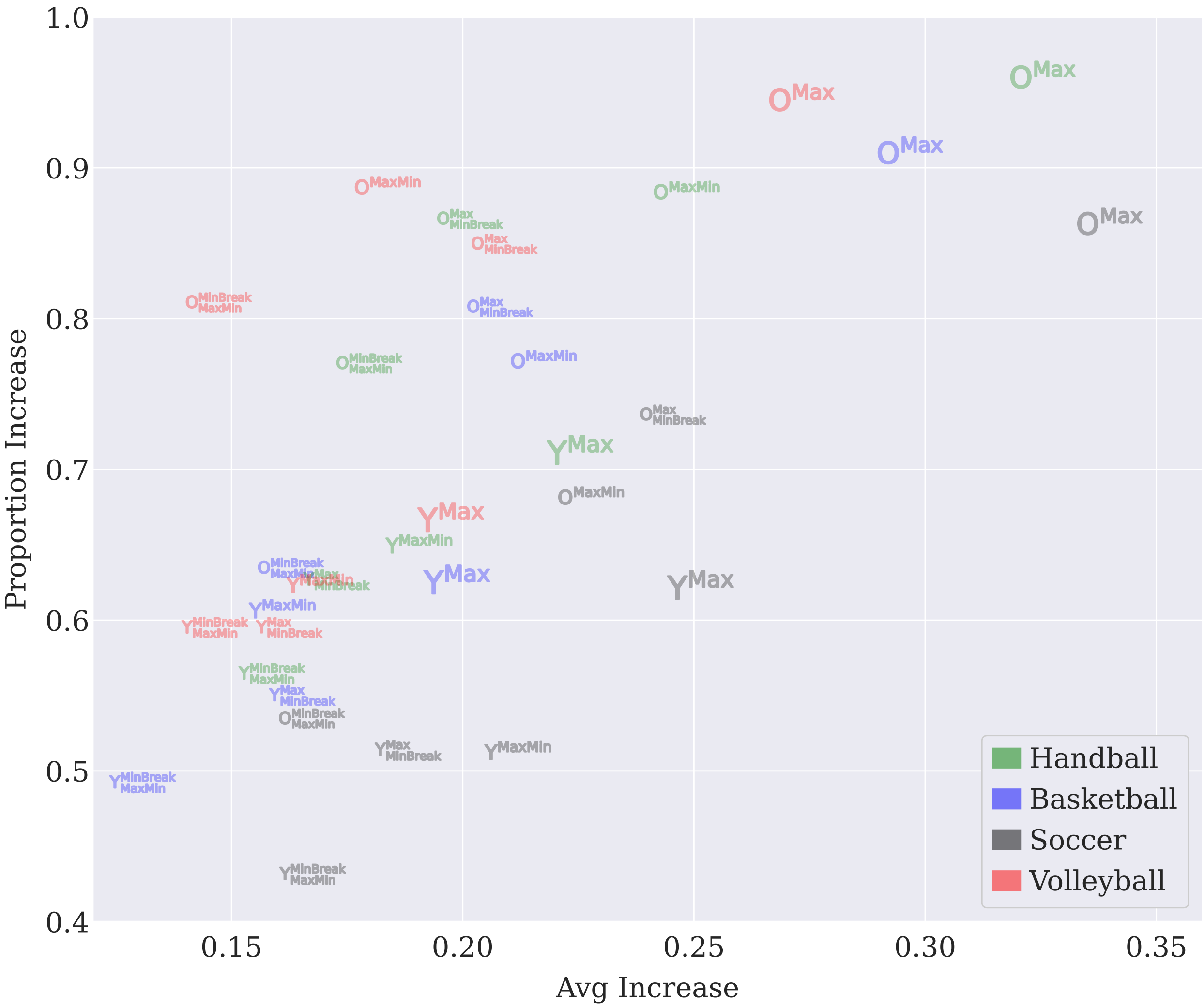

Figure D.3: Results for **REC**.

Appendix E. The Impact of Strength of Schedule Balance on Tournament Efficacy and Stadium Attendance

Competitive balance is widely recognized as a key determinant of demand in sports leagues, as it shapes fans' engagement and the perceived attractiveness of tournaments. While previous studies have examined competitive balance from various perspectives, less attention has been given to how the sequencing of matches affects either perceived or actual balance throughout a season. In this section, we introduce the Strength of Schedule Balance (SSB) metric, which captures how evenly the difficulty of opponents is distributed across a team's schedule. Using 22 seasons of English Premier League data, we investigate whether unbalanced schedules—where teams predominantly face strong or weak opponents early on—impact two central outcomes: (i) tournament efficacy, defined as the tournament's ability to rank teams according to their true strengths, and (ii) stadium attendance, as a measure of fan engagement. Our analysis shows no statistically significant evidence that unbalanced schedules affect final standings or attendance levels. These results suggest that while scheduling influences the sequence of competition, it does not materially alter tournament fairness or demand in the context of long round-robin leagues.

Appendix E.1. Measuring Strength of Schedule Balance

In a single round-robin tournament, let $R = \{r_1, \ldots, r_n\}$ represent the set of teams, ordered by their skill levels such that r_1 is the strongest team and r_n is the weakest. For practical purposes, we adopt an *oracle* approach, inferring team skill directly from their final standings in the tournament.

For each team r_i, its schedule, denoted as $\mathbf{S}_i$, is an ordered list of all opponents faced throughout the tournament. As each team in a single round-robin tournament faces every other team exactly once, $\mathbf{S}_i$ is effectively a permutation of R_{-i}, where R_{-i} denotes the ordered list R excluding team r_i.

We introduce the *Strength of Schedule Balance* metric to quantify how opponent difficulty is distributed across a team's schedule. A schedule is considered **balanced** if the skill levels of opponents faced are relatively uniform throughout the tournament rounds. Conversely, a schedule is deemed **unbalanced** if a team predominantly faces stronger opponents early on and weaker opponents later, or vice-versa.

More formally, we define the Strength of Schedule Balance, $B(\mathbf{S}_i)$, for team r_i as the *Spearman's rank correlation coefficient* between its schedule $\mathbf{S}_i$ and R_{-i}. Here, R_{-i} represents a perfectly ordered sequence of opponents from strongest to weakest, serving as the benchmark for an extremely unbalanced schedule (specifically, one where difficulty consistently decreases).

In summary, $B(\mathbf{S}_i)$ can be interpreted as follows:

- When $B(\mathbf{S}_i) = 1$, team r_i faced the most **unbalanced and progressively weakest** schedule possible. This means it consistently encountered opponents in descending order of strength (e.g., strongest opponent first, followed by progressively weaker ones).

- When $B(\mathbf{S}_i) = -1$, team r_i faced the most **unbalanced and progressively strongest** schedule possible. This indicates it consistently encountered opponents in ascending order of strength (e.g., weakest opponent first, followed by progressively stronger ones).

- When $B(\mathbf{S}_i) \approx 0$, the strength of the schedule faced by team r_i is considered **balanced** throughout the tournament, implying a relatively uniform distribution of opponent skill.

Appendix E.2. Data

To assess the impact of Strength of Schedule Balance on tournament efficacy and stadium attendance, we gathered data from the English Premier League from the website *Football Web Pages*[1]. Our dataset spans from the 2001/2002 season to the 2024/2025 season, specifically excluding the 2019/2020 and 2020/2021 seasons due to disruptions caused by the COVID-19 pandemic.

For each of these 22 seasons, we collected the final league tables to establish the ranking of teams R. We also compiled the complete schedules for all participating teams, resulting in a total of 440 individual team schedules. Finally, we obtained the stadium attendance figures for every match.

Appendix E.3. Methodology

For the subsequent sections, we address a central question: **Does the Strength of Schedule Balance influence tournament efficacy and stadium attendance?** Our hypothesis posits that it has no discernible effect on **tournament efficacy** (detailed in Section Appendix E.4) or **stadium attendance** (discussed in Section Appendix E.5). To investigate this, we first computed the Strength of Schedule Balance ($B(\mathbf{S}_i)$) for *the first half* of all 440 schedules within our dataset. We considered only the first half to emulate a single round-robin tournament, in which all teams face each other exactly once. These schedules were then categorized into three distinct, non-overlapping groups based on their $B(\mathbf{S}_i)$ values, allowing us to observe variations in tournament efficacy and stadium attendance across these categories:

- G_0: **Balanced schedules**, where $-0.3 \leq B(\mathbf{S}_i) \leq 0.3$.

[1] https://www.footballwebpages.co.uk/

- G_{-}: **Unbalanced weak schedules**, where $B(\mathbf{S}_i) < -0.3$.
- G_{+}: **Unbalanced strong schedules**, where $B(\mathbf{S}_i) > 0.3$.

For illustration, consider the 2003/2004 season, where Aston Villa finished 6th. Their schedule, denoted as $\mathbf{S}_6 = \{13, 4, 1, 18, 16, 7, 2, 8, 10, 17, 5, 11, 14, 12, 3, 20, 15, 19, 9\}$, yielded a Strength of Schedule Balance $B(\mathbf{S}_6) = 0.3053$. Conversely, in the 2024/2025 season, Leicester City finished 18th. Their schedule, $\mathbf{S}_{18} = \{17, 11, 6, 12, 13, 2, 9, 20, 7, 19, 15, 4, 10, 14, 8, 5, 16, 1, 3\}$, resulted in $B(\mathbf{S}_{18}) = -0.3053$. Even though the magnitudes of these correlations are not high, these schedules indicate a moderate level of unbalance.

Table E.2 details the number of schedules in each group and their average Strength of Schedule Balance ($\overline{B}(\mathbf{S}_i)$). Observe that while most schedules are balanced, a significant number are unbalanced. The most unbalanced strong schedule was faced by Wolverhampton Wanderers in the 2024/2025 season, who finished 16th with a schedule $\mathbf{S}_{16} = \{2, 4, 7, 5, 6, 1, 10, 3, 8, 12, 20, 11, 9, 13, 14, 19, 18, 15, 17\}$, resulting in $B(\mathbf{S}_{16}) = 0.83$. Conversely, the most unbalanced weak schedule belonged to Southampton in the 2017/2018 season, who finished 17th with a schedule $\mathbf{S}_{17} = \{18, 13, 16, 14, 11, 2, 19, 10, 20, 15, 7, 4, 8, 1, 12, 6, 9, 5, 3\}$, yielding $B(\mathbf{S}_{17}) = -0.60$.

Group	**n**	$\overline{B}(\mathcal{S}_i)$
G_{-1}	42	-0.41
G_0	360	0.00
G_{+1}	38	0.41

Table E.2: Number of schedules (n) and average Strength of Schedule Balance ($\overline{B}(\mathcal{S}_i)$) per group.

Following this grouping, we conducted the standard Mann-Whitney statistical significance test to ascertain if the observed values among these groups are statistically distinct. We report the p-values for each paired comparison between groups. Additionally, we provide graphical illustrations to depict the distribution of values within each group.

Appendix E.4. Impact on Tournament Efficacy

Tournament efficacy is measured as the propensity of a tournament to rank teams according to their strengths. Therefore, if unbalanced schedules offer advantages or disadvantages, then the expected rank of the teams under these schedules should be different from the ones under balanced schedules.

Figure E.4 presents a boxplot of the final ranks achieved by teams in each group. The plot shows that teams with unbalanced strong schedules, who face the best opponents early in the tournament, tend to finish with slightly lower ranks than those with balanced schedules. Conversely, teams facing unbalanced weak schedules, who play against the worst opponents early on, appear to finish with slightly higher ranks. These results suggest that an early schedule against top-tier teams offers a mild advantage, while an early schedule against lower-tier teams provides a slight disadvantage, compared to a balanced schedule.

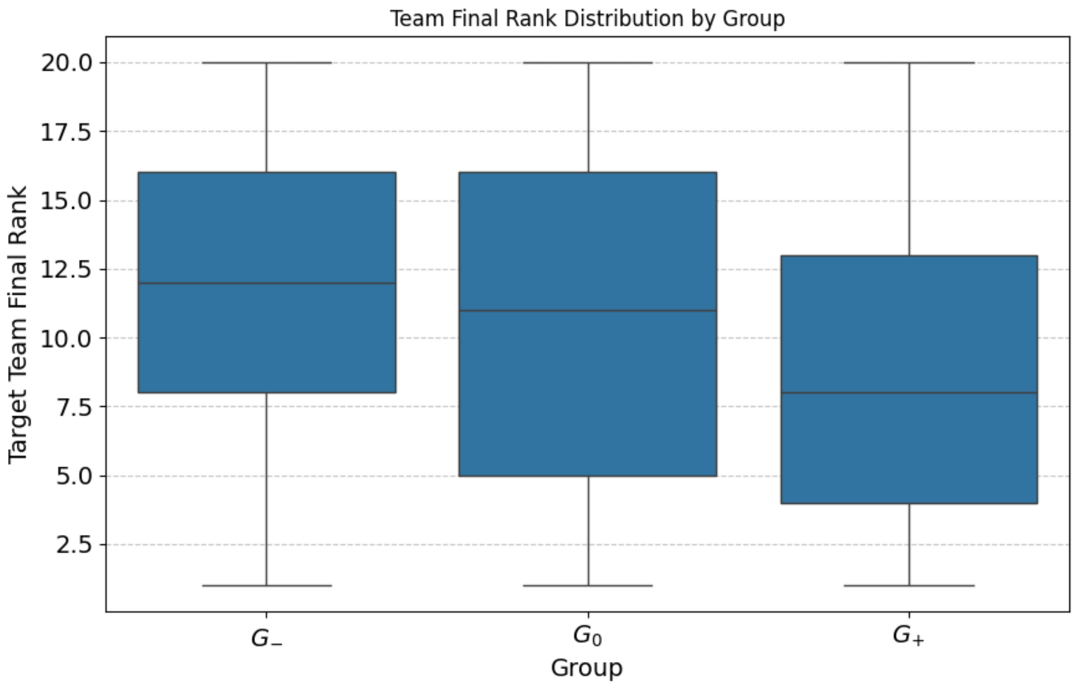

Figure E.4: Box-plot of the final ranks achieved by teams of each group.

To test the statistical significance of these findings, we performed a *Mann-Whitney U test* for every pair of groups. This test allowed us to determine if the final rank distributions of the groups are statistically different from each other. The p-values for each paired test are reported in Table E.3. As the table shows, all of the p-values are above the standard significance level of 0.05. This lack of a statistically significant result means we do not have strong enough evidence to reject the null hypothesis, which states that the expected final ranks of teams are the same regardless of whether they face unbalanced schedules.

Group	p-value vs G_-	p-value vs. G_0	p-value vs G_+
G_-	1.0	0.32	0.06
G_0	0.32	1.0	0.17
G_+	0.06	0.17	1.0

Table E.3: P-values for the Mann-Whitney statistical significance test comparing whether the distribution of final ranks between pairs of groups are different.

Appendix E.5. Impact on Stadium Attendance

For each team i and season y, we measure stadium attendance as the average occupancy of its home games. Because the maximum capacities of the stadiums used by the 44 teams varied over the 22 seasons analyzed, we adopted an approximate heuristic to estimate these capacities. Specifically, for each team i and season y, we assumed that the stadium's maximum capacity corresponded to the highest recorded attendance in a home game that season. The average occupancy was then computed by dividing the mean home game attendance by this maximum attendance.

Figure E.5 presents a boxplot of the average home game occupancy achieved by teams in each group. The plot reveals no apparent differences among the groups, except for a larger variance and a greater number of outliers in the balanced schedules group (G_0). This outcome is expected, as the number of schedules in this group is nearly ten times greater than in the other groups. Additionally, we observe that, across all groups, most average occupancies exceed 95%. These results suggest that unbalanced schedules have minimal impact on stadium occupancy.

As in the previous section, we further investigate this hypothesis by applying the Mann–Whitney statistical significance test to every pair of groups, in order to determine whether their distributions of average home game occupancy differ significantly. The resulting p-values for each pairwise comparison are reported in Table E.4. As shown in the table, all p-values are well above the standard significance threshold of 0.05. This lack of statistically significant differences indicates that we do not have sufficient evidence to reject the null hypothesis — that is, the Strength of Schedule Balance does not appear to affect stadium occupancy.

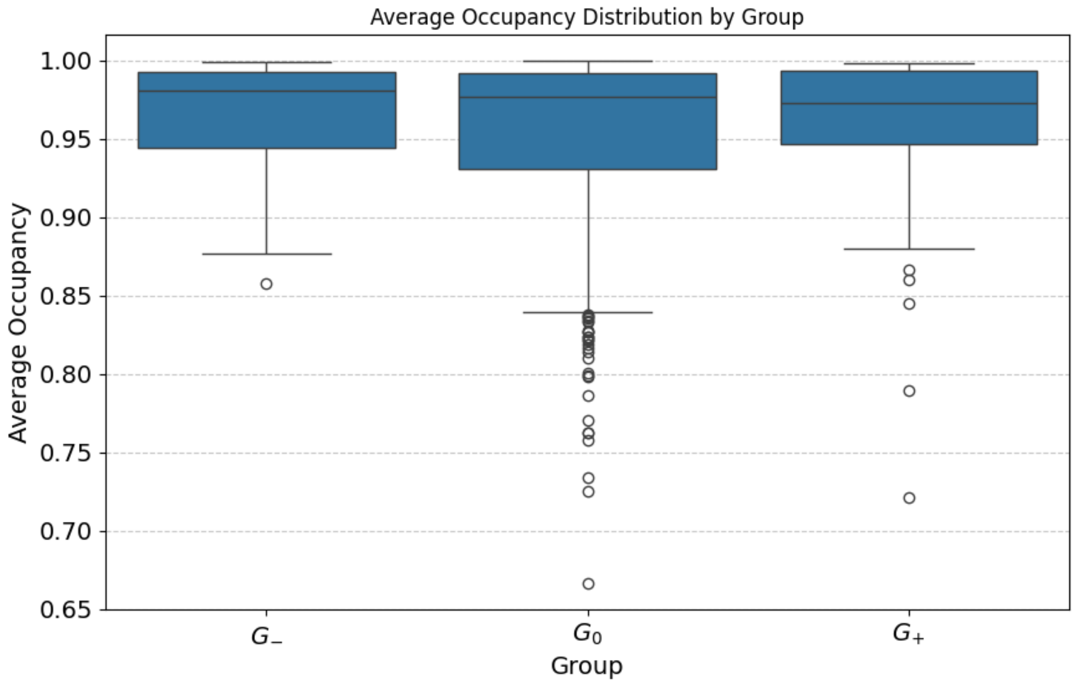

Figure E.5: Boxplot showing the distribution of average home game occupancy for teams within each group.

Group	**p-value vs G_-**	**p-value vs. G_0**	**p-value vs G_+**
G_-	1.0	0.52	0.87
G_0	0.52	1.0	0.48
G_+	0.87	0.48	1.0

Table E.4: P-values for the Mann-Whitney statistical significance test comparing whether the distribution of average home game occupancies between pairs of groups are different.

Appendix E.6. Discussion

Our results indicate that while some descriptive differences emerge (e.g., teams facing stronger opponents early in the tournament tend to finish with slightly lower ranks), these effects are not statistically significant. Across all comparisons, the Mann–Whitney tests fail to reject the null hypotheses, showing no systematic relationship between SSB and final league standings. Similarly, stadium attendance appears unaffected by schedule imbalance: average occupancy levels remain consistently high across all groups, and statistical tests confirm no significant differences.

These findings suggest that unbalanced schedules, as captured by SSB, do not materially influence either the competitive outcomes of tournaments or fan attendance. From a practical standpoint, this implies that organizers have flexibility in scheduling without risking measurable impacts on fairness or demand, at least within the context of long tournaments such as the English Premier League.

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